Hongzheng Tian

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Education

- **University of California, Irvine (UCI), Irvine, CA, USA** Sep. 2022 – Jun. 2027 (expected)

  Ph.D. in Computer Engineering **Advisor:** Prof. Sitao Huang

- **University of California, Irvine (UCI), Irvine, CA, USA** Sep. 2020 – Jun. 2022

  M.S. in Electrical and Computer Engineering

- **University of California, Riverside (UCR), Riverside, CA, USA** Aug. 2019 – Jun. 2020

  UCR Graduate Preparation Program in Engineering (UCR-GPP-E), one-year exchange program

- **University of British Columbia (UBC), Vancouver, BC, Canada** Jul. 2018 – Aug. 2018

  Vancouver Summer Program in Communication Systems and Digital Systems Design with FPGA

- **Northwestern Polytechnical University (NWPU), Xi'an, Shaanxi, China** Sep. 2016 – Jun. 2020

  B.Eng. in Microelectronics Science and Engineering

Research Experience

- **CoVA: Compiler for Versatile Architectures (Ph.D. Thesis)** Jun. 2023 – Present

  CORSA Lab, UCI; initiated during the 2023 HPE internship
  Advisor: Prof. Sitao Huang **Mentor:** Dr. Alok Mishra (Hewlett Packard Labs) **Role:** Sole developer

  - Built an MLIR-based compiler for NumPy programs: a custom dialect (38 ops) and 30 passes on top of upstream bufferization, lowering to six CPU/GPU code-generation routes (LLVM, OpenMP CPU/GPU, EmitC C++, and NVVM).
  - On CPU, achieved a 3.5× geomean speedup over NumPy across 53 NPBench workloads, outperforming Numba and DaCe; 53/54 CPU and 52/54 GPU workloads pass strict golden-reference checks.
  - Designed region-level storage and alias analyses that prove loop independence and keep buffers GPU-resident across control flow; implemented GPU thread-block mapping and deterministic reductions whose results are independent of thread count.
  - Implemented stream and scratch-buffer reuse and analysis-guided segmented synchronization in the CUDA runtime.
  - Built a PyTorch path that packages models into persistent `.pt2` executables through torch.export and AOTInductor, with verified Triton pattern specialization; runs end-to-end GPT-2 generation.

  Publication: SC24 ACM Student Research Competition Poster [S4]

- **HeteroTask: Distributed Heterogeneous Task Runtime** Jan. 2025 – Apr. 2026

  CORSA Lab, UCI **Role:** Sole developer

  - Built a DAG execution framework with task-variant profiling, memory-aware device and variant selection, communication-cost modeling, and buffer lifetime planning.
  - Implemented a C++/MPI runtime that executes planned task placements, dependency transfers, and buffer allocation and release across CPU/GPU nodes; its scheduling concepts later informed CoVA.

- **PyAIE: A Python-based Programming Framework for Versal ACAP** Sep. 2022 – Jun. 2023

  CORSA Lab, UCI **Advisor:** Prof. Sitao Huang **Role:** Led design and implementation

  - Built a Python-based framework that generates AI Engine kernels (with AIE vector intrinsics), programmable-logic and host code, and configuration scripts for AMD Versal ACAP.

  Publications: DAC 2023 Late Breaking Results [S2]; LATTE 2023 [S1]

• A Sparsity-Aware Autonomous Path Planning Accelerator with Algorithm-Architecture Co-Design Jan. 2023 – Sep. 2023

Joint project of the CORSA Lab, UCI and the School of Cyberspace Science and Technology, Beijing Institute of Technology
Advisors: Prof. Sitao Huang, Prof. Yanjun Zhang, and Dr. Bo Yu

- FPGA-based acceleration of autonomous path planning, focusing on sparse matrix-vector multiplication and preconditioned conjugate gradient (PCG) computation for quadratic programming.

Publications: ACM TACO 2025 [J2]; ICCAD 2024 [C1]; FPGA 2024 Poster [S3]

• SAMT: Optimized Spatial Architecture Mapping Flow for Transformer Accelerators

CORSA Lab, UCI

- A framework for optimizing dataflow mappings of Transformer inference workloads on spatial accelerators, combining mapping search with dynamic operator fusion to improve latency and energy efficiency across edge, mobile, and cloud platforms.

Publication: arXiv:2410.07407, 2024 [P1]

• Design Exploration of Associative Processor Implementation on FPGA Apr. 2021 – Feb. 2022

Center for Embedded and Cyber-physical Systems (CECS), UCI
Advisors: Prof. Fadi J. Kurdahi, Dr. Mohammed E. Fouda, and Dr. Minjun Seo

- Designed and implemented an FPGA-based associative processor for in-memory computing, including content addressable memory and supporting circuits (controller, data and instruction caches, and program counter).

Publication: IEEE TCAS-II 2023, presented at ISCAS 2023 [J1]

Industry Experience

• Research Associate Intern, Hewlett Packard Enterprise (Hewlett Packard Labs), Milpitas, CA Jun. 2024 – Sep. 2024

HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems **Role:** Lead developer
Jointly supervised by Prof. Sitao Huang (CORSA Lab, UCI) and Dr. Alok Mishra (Hewlett Packard Labs); started at UCI in Mar. 2024 and completed during the internship

- Led development of a benchmark suite of 11 multi-kernel applications spanning image processing, machine learning, numerical computation, and physical simulation, with Python, OpenMP, CUDA, and HLS implementations for CPUs, GPUs, and FPGAs.
- Built a JSON-configured Python driver with Jinja-generated Makefiles for backend-specific builds, benchmark execution, profiler integration, and experiment logging.
- Best Paper Award – Artifact Track at ICPE 2025 [C2]; open-sourced as HewlettPackard/HeteroBench and listed in SPEC RG Tools.

• Research Associate Intern, Hewlett Packard Enterprise (Hewlett Packard Labs), Milpitas, CA Jun. 2023 – Sep. 2023

CoVA: Compiler for Versatile Architectures

- Built CoVA's initial compilation framework: a custom MLIR dialect separating high-level program semantics from target-specific lowering, MLIR-to-LLVM lowering, and Python bindings.
- Implemented compiler analysis and transformation passes for the heterogeneous compilation flow; the project continued as my Ph.D. thesis.
- This work led to a U.S. patent application with Hewlett Packard Enterprise, now allowed [PT1].

• Development Engineer Trainee, IMBED LLC Sep. 2021 – Jan. 2022

MQ Decoder for a Real-Time FPGA-based JPEG2000 Player for Digital Cinema

- Implemented the JPEG2000 decoder in C according to the official JPEG2000 specification.
- Designed and simulated the tier-1 entropy decoder (MQ decoder) in Verilog HDL, drawing on published MQ-decoder improvements.
- Optimized the decoder to meet timing and throughput constraints, reaching up to 450 MHz.

Publications

Conference Papers

[C2] **H. Tian**, A. Mishra, Z. Chen, R. P. Hong Enriquez, D. Milojicic, E. Frachtenberg, and S. Huang. "HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems." *Proceedings of the 16th ACM/SPEC International Conference on Performance Engineering (ICPE)*, pp. 320–333, 2025. **Best Paper Award – Artifact Track.**

[C1] Y. Zhang, X. Niu, Y. Zhang, **H. Tian**, B. Yu, S. Liu, and S. Huang. "A Sparsity-Aware Autonomous Path Planning Accelerator with Algorithm-Architecture Co-Design." *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2024.

Journal Articles

[J2] Y. Zhang, X. Niu, **H. Tian**, Y. Zhang, B. Yu, S. Liu, and S. Huang. "A Sparsity-Aware Autonomous Path Planning Accelerator with HW/SW Co-Design and Multi-Level Dataflow Optimization." *ACM Transactions on Architecture and Code Optimization (TACO)*, 22(3):1–25, 2025.

[J1] **H. Tian**, M. E. Fouda, M. Seo, and F. J. Kurdahi. "FPGA Implementation of Associative Processors." *IEEE Transactions on Circuits and Systems II: Express Briefs*, 70(5):1774–1778, 2023. Presented at ISCAS 2023.

Short Papers, Posters, and Workshop Papers

[S4] **H. Tian**, A. Mishra, and S. Huang. "CoVA: Compiler for Versatile Architectures." *ACM Student Research Competition Poster, SC24*, 2024.

[S3] X. Niu, Y. Zhang, Y. Zhang, **H. Tian**, B. Yu, S. Liu, and S. Huang. "Accelerating Autonomous Path Planning on FPGAs with Sparsity-Aware HW/SW Co-Optimizations." *ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA)*, poster, 2024.

[S2] **H. Tian**, S. Yang, Y. Cha, and S. Huang. "Late Breaking Results: PyAIE: A Python-based Programming Framework for Versal ACAP Platforms." *60th ACM/IEEE Design Automation Conference (DAC)*, pp. 1–2, 2023.

[S1] **H. Tian**, S. Yang, Y. Cha, and S. Huang. "PyAIE: A Python-based Compiler Framework for Versal ACAP Platforms." *Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE)*, 2023.

Preprints

[P1] H. Xu, F. Tahmasebi, Y. Qiao, **H. Tian**, H. Kwon, and S. Huang. "Optimized Spatial Architecture Mapping Flow for Transformer Accelerators." *arXiv:2410.07407*, 2024.

Patents

[PT1] A. Mishra, **H. Tian**, R. P. Hong Enriquez, M. P. Dziubinski, X. Zhan, and D. S. Milojicic. "Microarchitectural-Neutral Automatic Type and Shape Inference and Cross-Microarchitecture Invocation." U.S. Patent Application No. 18/678,875, filed May 30, 2024; published as US 2025/0370738 A1. **Allowed** (Notice of Allowance, 2026). Assignee: Hewlett Packard Enterprise.

Honors & Awards

- **Best Paper Award – Artifact Track, ICPE 2025** — May 2025

  For *HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems*

- **Nominee, EECS Graduate Student of the Year Award, UC Irvine** — Mar. 2024
- **Second-class Scholarship for Outstanding Students, NWPU** — Dec. 2017 & Dec. 2018
- **Second Prize, 19th Undergraduate Mathematical Contest in Modelling, NWPU** — Jun. 2018

Teaching

- **Teaching Assistant, EECS 112L Organization of Digital Computers Laboratory, UC Irvine** — Fall 2026
- **Teaching Assistant, EECS 112 Organization of Digital Computers, UC Irvine** — Fall 2023

Presentations & Activities

- **ICPE 2025 "Benchmarking in the Data Center: Expanding to the Cloud" Workshop.** Invited session chair and speaker; declined due to travel constraints.
- **SC24 ACM Student Research Competition.** Poster presentation, Atlanta, GA, 2024.
- **UCI GEECS Showcase.** Poster presentation, Irvine, CA, 2024.

- **Young Fellow Program, 60th Design Automation Conference.** Poster presentation, San Francisco, CA, 2023.
- **LATTE '23, Workshop on Languages, Tools, and Techniques for Accelerator Design.** Presentation of PyAIE [S1], 2023.

Skills

Programming: C, C++, Python, Verilog HDL, VHDL, MATLAB

Compilers & ML systems: MLIR, LLVM/NVVM, OpenMP offloading, EmitC; PyTorch (torch.export, AOTInductor), Triton, torch-mlir

Parallel computing: OpenMP, MPI, CUDA, high-level synthesis (Vitis HLS)

EDA & tools: AMD/Xilinx Vivado and Vitis, Synopsys, Cadence, Quartus, ModelSim

Undergraduate Projects

• Digital Multi-Channel Lock-in Amplifier Based on FPGA (Senior Design) Sep. 2019 – Mar. 2020

Bourns College of Engineering, UCR **Advisor:** Prof. Shane Cybart

- Co-designed an FPGA-based multi-channel lock-in amplifier that extracts a signal with a known carrier wave from an extremely noisy environment.
- Designed the digital phase-locked loop that synchronizes to the external reference signal, the internal square-wave reference generator, and the vector calculator for the IIR filter outputs.
- The final system extracts a 0.5 μV sinusoidal signal from 500 mVpp white noise.

• Convolutional Neural Network (CNN) Based on FPGA Sep. 2019 – Dec. 2019

Bourns College of Engineering, UCR **Advisor:** Prof. Philip Brisk

- Assisted in designing an FPGA-based CNN for processing medical images; studied CNN fundamentals and the lab's C++ CNN implementation.
- Used Vivado HLS to translate the C++ design to Verilog HDL and packaged it as an IP core for further tests and experiments.

• Q-Learning for the Maze Path-Finding Problem (EE XRC 260 Course Project) May 2020 – Jun. 2020

UCR **Instructor:** Prof. Sheldon Tan

- Studied Markov decision processes and (deep) Q-learning, and implemented Q-learning in MATLAB to find the shortest path out of a maze.

• Undergraduate Mathematical Contest in Modelling, NWPU (Second Prize) May 2018

- Designed a model using cluster analysis and hierarchical analysis of a text's topological and logical structure, and used MATLAB character-frequency statistics to determine whether a book was written by a single author.

Extracurricular Activities

• Theory Department Undersecretary & Photography Team Leader, Astronomical Society, NWPU Aug. 2017 – Jun. 2018

- Organized astronomy lectures, competitions, and night-sky photography field trips; served as question setter and judge for the Astronomical Joint Competition for Universities and Colleges in Shaanxi Province.

• Learning Department Secretary, Student Union of the School of Software and Microelectronics, NWPU Aug. 2017 – Jun. 2018

- Organized academic lectures, study-experience meetings, subject competitions, and debate contests; helped students resolve coursework questions; set, proctored, and graded an advanced mathematics exam.